

MULTI-DIMENSIONAL PROSODY JUDGMENT FOR LIVESTREAM SPEECH SYNTHESIS

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ABSTRACT

We build an automatic prosody evaluation model for e-commerce livestream speech synthesis, with pairwise evaluation and Best-of-8 candidate selection as its current applications. We compare against SpeechJudge-GRM and Gemini-3.1-pro-preview and use Qwen3-Omni as the student backbone. We develop a pipeline combining Gemini distillation, two-stage supervised fine-tuning, and GRPO reinforcement learning. Across four evaluation sets and a unanimous subset of T1, Live-SpeechJudge+GRPO with 10-sample balanced-order aggregation, which averages five judgments in each A/B presentation order, has higher point accuracy than a single Gemini judgment in all reported columns. On a high-confidence subset of 136 out of 400 Best-of-8 candidate sets, its tournament-selected candidate achieves Hit@1/2/3 rates of 72.06%, 77.94%, and 85.29%, respectively. We further study verdict coupling, a pattern in which per-dimension verdicts follow the overall verdict. To remove the shared overall-verdict target, we query the teacher separately for each of the four core dimensions C1–C4 and train Decouple-Live-SpeechJudge without an overall conclusion. Its non-unanimous outputs show that the model can issue different verdicts across dimensions. In one fixed-seed run, per-dimension GRPO changes pooled point agreement from 73.64% to 77.94% with one sample and from 84.10% to 86.10% with 10-sample balanced-order aggregation. At 10 samples, however, paired-bootstrap intervals for the individual dimensions include zero, so these changes are descriptive rather than statistically significant.

Index Terms— prosody evaluation, expressive speech evaluation, LLM-as-a-judge, reinforcement learning

1. INTRODUCTION

E-commerce livestream TTS must jointly satisfy fluency, varied intonation, plausible emotion, high energy, emphasis on sales cues, and smooth switches between product explanation and audience interaction. A single quality score cannot represent these requirements. We therefore seek a pairwise evalu-

ator that provides fine-grained prosodic feedback for two applications: selecting the best of N sampled utterances at serving time and supplying scalable rewards for TTS preference optimization [1, 2].

Reference-free MOS predictors [3, 4] regress a single naturalness score and do not represent the expressive and interaction-mode phenomena in our rubric. SpeechJudge-GRM [6] is not specialized for livestream speech and exhibits a large B-position gap under our protocol, while the stronger Gemini-3.1-pro-preview [12] is too costly for our deployment setting. We therefore distill it into the audio-capable Qwen3-Omni [14].

A second problem is **verdict coupling**: per-dimension score differences from the teacher, SpeechJudge-GRM, and our coupled student often follow the overall winner. A multi-dimensional rubric can then collapse to one bit and fail to provide varied rewards. We address this by removing the overall-verdict target and applying dimension-specific supervision.

This paper makes three contributions.

1. A livestream audio evaluation benchmark. We introduce a seven-dimension pairwise rubric and 1043 human-annotated pairs spanning 114 speakers, cross-model, same-model, and human-vs.-TTS comparisons, three speaking styles, and three duration bands.

2. Live-SpeechJudge. We combine 4-way swap-consistency filtering, a two-stage curriculum, dimension-weighted SFT, and GRPO. Its 10-sample balanced-order aggregation has higher point accuracy than one Gemini judgment on all reported columns. The model also shows a small measured presentation-slot gap and obtains 72.06%/77.94%/85.29% Hit@1/2/3 on the retained high-confidence Best-of-8 sets.

3. Decouple-Live-SpeechJudge (D-LSJ). D-LSJ queries supervision separately for C1–C4, masks uncertain pair-dimensions, removes the shared overall verdict, and applies each GRPO advantage only to its dimension’s rationale span. It produces non-unanimous verdicts on 24.3–60.6% of evaluated pairs. In one fixed-seed run, span-local GRPO changes pooled 10-sample human agreement from 84.10% to 86.10%; the per-dimension confidence intervals include zero.

Work done at the Alibaba TaoBao & TianMao Group Livestream AIGC team, which supported this research.

2. RELATED WORK

2.1. Automatic speech quality assessment

Reference-free MOS predictors [3, 4, 5] regress a scalar — or a fixed set of transmission-quality scalars — trained to match crowdsourced ratings of largely read speech. They are cheap enough to run inside an RL loop, but they emit no rationale, they score a clip in isolation rather than comparing two renderings of the same text, and they are weakly correlated with human judgments of expressiveness. The long-form benchmark SwanBench-Speech [10] quantifies this directly: over 1101 samples spanning 17 scenarios, modern TTS already approaches human level on computable acoustic metrics such as timbre consistency and content accuracy, while prosodic coherence and expressive richness remain clearly behind, and conventional MOS networks agree with human expressiveness ratings far worse than a large audio-language judge does. That benchmark consequently has to delegate exactly those subjective dimensions to SpeechJudge [6] and Gemini. A scalar naturalness regressor is therefore not a candidate for our task, neither as a best-of- N comparator nor as a per-dimension reward.

2.2. Model-as-a-judge benchmarks for speech synthesis

A parallel line uses a strong audio LLM as the evaluator and invests the effort in the test material instead. EmergentTTS-Eval [9] grows 1645 test cases from a small pool of human-written seed prompts by iteratively prompting an LLM to make each case harder, organized into six challenge categories — emotions, paralinguistics, foreign words, syntactic complexity, complex pronunciation, and questions — and scores systems with a large audio-language model as judge, reporting high correlation with human preference. SwanBench-Speech does the same at minute-scale duration and 17 scenarios. These works establish that model-as-a-judge is a viable evaluation protocol for expressive TTS, and they motivate our rubric design, but they are consumers of an expensive closed judge rather than builders of a cheap one, and they produce a benchmark score for a system rather than a per-utterance reward vector that an RL loop can differentiate against. Our test sets also differ in construction intent: rather than curating adversarial text, each pair holds the transcript fixed, reducing lexical-content confounding and focusing the comparison on differences in the rendered delivery.

2.3. SpeechJudge and generative speech reward models

SpeechJudge [6] is the closest system-level analogue to this work and our principal baseline. It contributes three artifacts around a single subjective axis, naturalness. SpeechJudge-Data is a human-feedback corpus of 99K speech pairs generated by a diverse set of advanced zero-shot TTS systems across several speech styles and languages, annotated for

both intelligibility and naturalness preference; from it the authors carve SpeechJudge-Eval, a benchmark on which they show that existing metrics and audio LLMs are weak — the strongest off-the-shelf model, Gemini-2.5-Flash, reaches under 70% agreement with human judgment. SpeechJudge-GRM is then a generative reward model initialized from Qwen2.5-Omni-7B and post-trained in two stages: supervised fine-tuning on chain-of-thought rationales, followed by GRPO reinforcement learning restricted to challenging cases. It attains 77.2% on SpeechJudge-Eval (79.4% with inference-time scaling at $N = 10$) against 72.7% for a classic Bradley-Terry reward model, and the authors demonstrate its use as a reward function when post-training a speech generation model. Its overall pipeline — a generative judge trained first with rationale supervision and then with RL — closely matches ours; we also tested our pipeline on the SpeechJudge backbone before migrating to Qwen3-Omni (Section 5.1).

Three differences separate it from what a livestream judge needs. First, *granularity*: SpeechJudge-GRM is natively trained to emit one naturalness preference and therefore does not directly supply the reward vector required by per-dimension TTS optimization. Under our common multi-dimensional prompt it can produce section-level assessments, which we use only to analyze whether those assessments follow the overall preference. Second, *domain and bias*: it is not specialized for e-commerce livestream speech, has low human-label agreement on our high-expressiveness sets, and carries a strong positional prior under our protocol, preferring whichever clip is presented second by +0.98 on a 10-point scale with a 66.8% B win rate (Table 4) — a bias of the kind first catalogued for text LLM judges [13]. Third, *supervision economics*: its accuracy rests on 99K human preference annotations, whereas our task-specific model uses roughly two thousand teacher-labeled pairs selected by 4-way swap consistency and ordered as a curriculum. Because the models, domains, and annotation sources differ, this is not a scale-controlled data-efficiency comparison with SpeechJudge; it instead shows that a comparatively small, confidence-filtered corpus is sufficient to obtain the reported point accuracy in our narrow deployment domain.

Two concurrent generative reward models extend the same idea. GSRM [7] decomposes naturalness into interpretable acoustic attributes followed by structured reasoning, is trained on 31K expert annotations plus a test set of real user interactions, and improves a speech LM when used as an RLHF reward. UniSRM [8] is the closest prior work to our *method*: it unifies four assessment tasks — single-utterance A/B preference, absolute multi-dimension scoring, scene-conditioned style consistency, and multi-turn dialogue judgment — in one Qwen2.5-Omni thinker trained by SFT and then GRPO, with a reward summing a format term, a final-answer accuracy term, and a reasoning-consistency term that compares the sign of each predicted per-dimension score difference against the teacher’s. It also removes position ar-

tifacts at the data level by shuffling A/B order and discarding cyclically inconsistent preferences. The difference from our method is where the per-dimension signal lands. UniSRM averages per-dimension agreement into a single scalar next to the final-answer term, so one group-level advantage is broadcast across the entire rationale and disagreeing dimensions can partially cancel. We instead remove the overall verdict from the training target and give each dimension its own group-normalized advantage on its rationale span, preventing direct cross-dimension cancellation in the token-level objective. Where a single preference is still required — for best-of- N selection, or to compare against a coupled judge — we recover one by weighting the per-dimension score differences, but that aggregation is an explicit rule applied outside the model rather than a verdict the rationale is written toward. This design produces non-unanimous core-dimension outputs on 24.3–60.6% of evaluated pairs, providing direct evidence of output diversity across dimensions.

2.4. Context- and dimension-aware judgment

CEAEval [11] argues that expressiveness is only meaningful relative to context and reframes the task as expressive *appropriateness*: whether a clip matches the communicative intent implied by its discourse-level narrative. It contributes CEAEval-D, 16.1 hours of Mandarin conversational speech drawn from a 3505-hour audiobook corpus and annotated along fifteen expressive dimensions, and CEAEval-M, which pairs a frozen text-only planner predicting an ideal expressive profile with a Speech-LLM judge trained in three stages — caption-based knowledge distillation, CoT scoring SFT, and GRPO on a rebalanced score distribution. We adopt their adaptive audio attention bias in Section 4.6; it trains stably but brings no clear accuracy gain in our setting. This null result does not establish whether the mechanism becomes beneficial at longer rationale lengths. Our task differs in two ways that matter for reward modeling: we compare two renderings of the *same* transcript rather than absolute-scoring a single clip against a text-derived ideal, and we require per-dimension verdicts that survive as an independent reward vector rather than a single appropriateness score.

3. TASK FORMULATION AND BENCHMARK

3.1. Pairwise multi-dimensional judgment

Let (a_A, a_B, x) be an instance, where a_A, a_B are two livestream audio clips rendering the same target transcript x . A judge π produces, for each evaluated rubric dimension d , a rationale c_d and an integer score pair $(s_d^A, s_d^B) \in \{1, \dots, 10\}^2$. The dimension verdict is

$$v_d = \text{sgn}(s_d^A - s_d^B) \in \{A, \text{Tie}, B\}, \quad (1)$$

with $v_d = \text{Tie}$ when the two scores are equal. Live-SpeechJudge evaluates all seven dimensions and then gen-

erates an overall section containing a 1–10 score for each clip, an overall verdict $V \in \{A, \text{Tie}, B\}$, and its own justification. Decouple-Live-SpeechJudge instead evaluates and emits only the four core dimensions C1–C4, with no L1–L3 sections and no overall verdict. When an aggregate comparison is required, we compute it externally from the four core score differences. Overall accuracy is the agreement of the generated or externally aggregated verdict with the human A/B label, and ties are counted as errors.

3.2. Rubric design

Table 1 lists the rubric. C1–C4 are always evaluated. L1–L3 are *activation-gated*: the judge first decides from x alone whether the dimension applies and, if not, emits a short “not activated” section with both scores 0, since forcing a verdict on an inapplicable dimension only injects noise into aggregation.

3.3. Test sets

We deliberately cover three difficulty regimes, since a judge that only separates human from TTS is useless for best-of- N .

- **T1** qwen3-TTS base vs. SFT, 417 pairs. **T1*** is the 119-pair subset on which all three annotators agree.
- **T2** qwen3-TTS-GDPO vs. itself (two independent sampling runs), 111 pairs, three-annotator unanimous. Hardest regime: both clips come from the same model, so the judge must distinguish quality variation among independent samples.
- **T3** qwen3-TTS-GDPO vs. BERT-CosyVoice [16], 315 pairs, reported as two annotation batches T3a (166) and T3b (149). Cross-model regime.
- **T4** human recording vs. qwen3-TTS, 200 pairs.

Construction is stratified rather than opportunistic. From a pool of 135 speakers we sample 114: 70 warm-recommendation, 27 passionate-promotion and 17 professional-explanation speakers. Transcripts are matched to each speaker’s category (passionate: snacks, womenswear, personal care, large and kitchen appliances; professional: phones, 3C, jewelry, gold; warm: baby, travel, beauty, cosmetics, tableware) and drawn evenly from ASR transcripts of real streams and from generated copy of three functional types (product entry, product introduction, order-pressing). Durations are balanced over 5–10 s, 10–20 s and 20–30 s. Before annotation, all evaluation partitions are frozen. No evaluation audio, audio pair, normalized transcript, speaker, or source livestream appears in the SFT/GRPO training or validation data.

Three annotators label every pair independently under a written protocol, blinded to system identity and audio source. They are Alibaba-contracted annotation staff who completed task-specific professional training. Human annotation uses

Table 1. Seven-dimension livestream prosody rubric (high-expressiveness profile). The gentle profile keeps C1–C3 with softened anchors and replaces C4 with “Gentle Style Fit” (energy control, steadiness, warmth, real-human calm-host feel).

ID	Dimension	Gate	Measured attributes
C1	Fluency & Naturalness	always	rate stability, phrasing, pause naturalness, coherence
C2	Intonation Variation	always	pitch range, sentence-final contour, stress pitch, inter-sentence contrast
C3	Emotional Expression	always	emotion-content match, intensity, warmth / appeal
C4	Livestream Expressiveness	always	energy, rate drive, rhythmic momentum, real-host feel
L1	Key Information Emphasis	text	acoustic prominence on price / discount / scarcity / call-to-action
L2	Emotional State Switching	text	capture of emotional turning points, transition smoothness
L3	Interaction Mode Switching	text	tone change at explanation-to-audience markers

the same rubric definitions, pairwise decision criteria and instructions as the Gemini prompt. Averaged across the main T1–T4 test sets, 45% of pairs receive a 3-0 decision and 55% receive a 2A1E or 2B1E decision, where E denotes an abstention; the final A/B label is determined by the non-abstaining majority. T1–T4 therefore evaluate the overall pairwise verdict under the multidimensional rubric; separate human tests for individual core dimensions are introduced in Section 5.5. Quality-control exclusions are limited to objective data problems such as corrupted audio, transcript mismatch, or protocol violations, and are made without access to model predictions.

4. METHOD

Figure 1 summarizes the system: one prompt, two rationale formats — coupled and decoupled — and a three-phase training pipeline built on swap-consistency-filtered teacher labels.

4.1. Teacher distillation with swap-consistency filtering

Human annotators provide pairwise preferences, but collecting detailed rationales at training scale is costly in our setting. We therefore distill Gemini-3.1-pro-preview. Naive distillation can propagate the teacher’s order sensitivity and low-confidence predictions. We therefore query the teacher four times per pair: twice in the presented order and twice with a_A and a_B swapped. Let k be the number of the four verdicts favoring the same clip after un-swapping. We keep $k = 4$ (“4-0”) as confident supervision and discard $k = 3$ and $k = 2$ (“2-2”) entirely. Disagreement across repeated and swapped queries indicates lower teacher confidence and may reflect order sensitivity, sampling variability, or intrinsic ambiguity. We un-swap the votes to identify the physical winner, then assign retained training pairs to balanced presentation orders so that the winner appears equally often in position A and position B. The supervision therefore does not systematically associate the winning label with either slot. Table 2 directly ablates this threshold using v0, which expands the training set to 2300 pairs by admitting both 4-0 and 3-1 outcomes while keeping the model, prompt, curriculum, optimization hyperparameters, and evaluation protocol identical to v1.

4.2. Two-stage curriculum SFT

The student is Qwen3-Omni. We attach LoRA [17] to attention projections only (q, k, v, o) , rank 32. Stage 1 trains on 820 qwen3-TTS-GDPO vs. human pairs — large-gap instances that teach the model what each dimension is. Stage 2 trains on 710 cross-model (vs. BERT-CosyVoice) and 600 same-model self-pairs, which teach fine discrimination. Table 2 compares this sequential schedule with stage-2-only training and with single-stage training on the merged data. After SFT, the rank-32 adapter is merged into the backbone weights.

4.3. Dimension-weighted token loss

We use per-section weights to control the contribution of each rubric dimension to supervised fine-tuning. Let T_d be the token indices of the section for dimension d in the target rationale, and α_d its SFT loss weight. We optimize

$$\mathcal{L}_{\text{SFT}}(\theta) = - \frac{\sum_d \alpha_d \sum_{t \in T_d} \log \pi_\theta(y_t \mid y_{<t}, a_A, a_B, x)}{\sum_d \alpha_d |T_d|}, \quad (2)$$

which is the usual next-token loss with per-section reweighting and a matching normalizer so that the effective learning rate does not drift with α_d . For the coupled model we use $\alpha = (1.5, 0.7, 1.5, 1.5, 1.5, 0.7, 0.7)$ over (C1..C4, L1..L3), plus 1.0 on the free-text summary and 2.0 on the conclusion line.

4.4. GRPO post-training

Error analysis on the SFT model shows that the remaining errors are mostly incorrect verdicts on difficult cases, suggesting that the preference judgment has not fully converged. This is exactly the regime where policy optimization against a verifiable reward can help, so we post-train with GRPO [2].

For each prompt q we sample a group of $G = 8$ completions $\{o_i\}$ from $\pi_{\theta_{\text{old}}}$. For a group-level scalar z_i , define

$$\mathcal{N}_G(z_i) = \frac{z_i - \text{mean}(\{z_j\}_{j=1}^G)}{\text{std}(\{z_j\}_{j=1}^G) + \epsilon}. \quad (3)$$

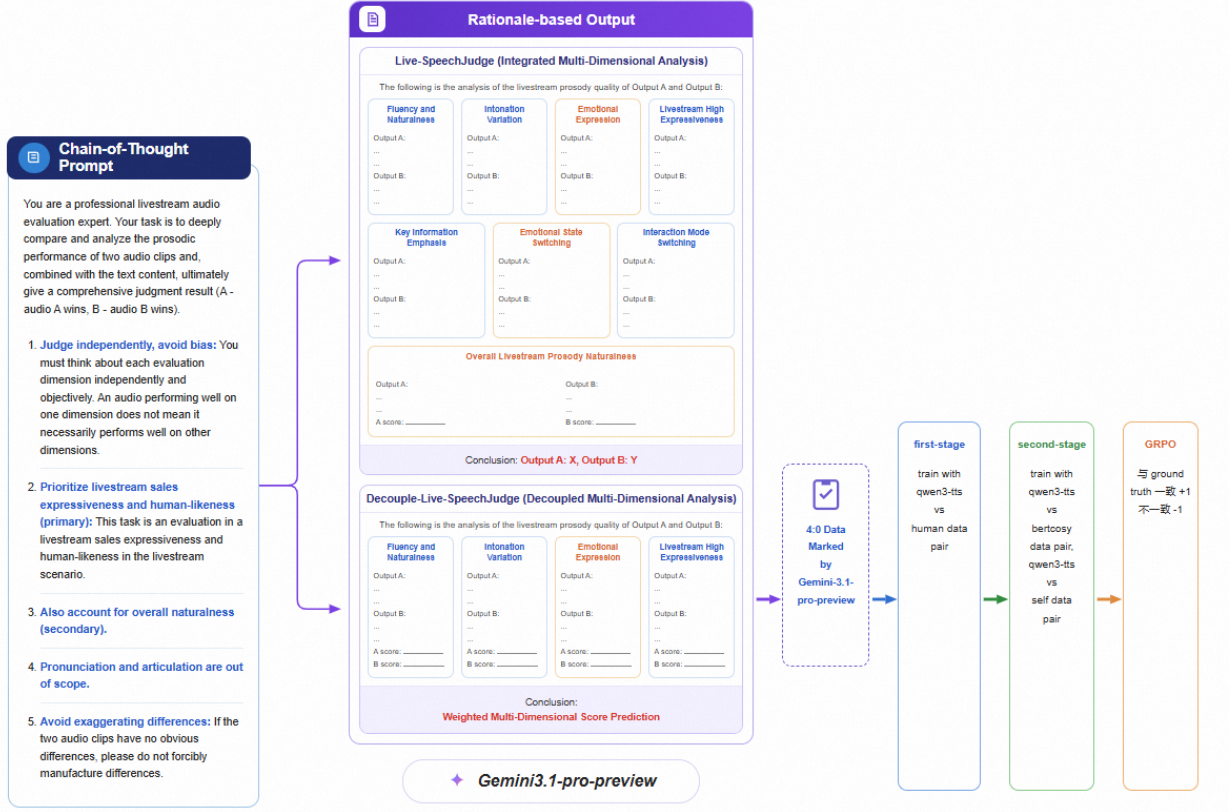


Fig. 1. Overview. **Left:** the chain-of-thought prompt shared by teacher and student, which fixes the rubric and states the independence, priority and no-exaggeration instructions. **Middle top:** the coupled judge Live-SpeechJudge emits one section per rubric dimension — four always-on plus three text-activated — and closes with an overall naturalness section and a single conclusion. **Middle bottom:** Decouple-Live-SpeechJudge emits only the four independently supervised core-dimension sections C1–C4 and no overall verdict; the reported preference is their weighted aggregate. **Right:** the training pipeline. For coupled SFT, Gemini-3.1-pro-preview labels each pair four times and only 4-0 swap-consistent pairs are retained; for decoupled SFT, each core dimension is queried separately and retained at 4-0, 3-1, or three agreeing non-tie votes plus one abstention. Stage 1 trains on qwen3-TTS vs. human pairs and stage 2 on cross-model and same-model pairs. Live-SpeechJudge GRPO uses one overall ± 1 agreement reward; Decouple-Live-SpeechJudge GRPO uses separate ± 1 rewards on the C1–C4 rationale spans.

Live-SpeechJudge uses one overall reward for each completion,

$$r_i^{\text{overall}} = \begin{cases} +1 & V_i = \tilde{V}, \\ -1 & \text{otherwise}, \end{cases} \quad (4)$$

where $\tilde{V}$ is the teacher’s overall verdict. Its advantage $\mathcal{N}_G(r_i^{\text{overall}})$ is applied to every generated token.

For Decouple-Live-SpeechJudge GRPO, each retained core dimension receives its own reward,

$$r_{i,d} = \begin{cases} +1 & v_{i,d} = \tilde{v}_d, \\ -1 & \text{otherwise}, \end{cases} \quad d \in \{C1, \dots, C4\}, \quad (5)$$

and is normalized independently within the rollout group. A dimension without a confident teacher A/B label is masked and contributes no advantage. Let $T_{i,d}$ be the token span of

dimension d in completion i , and let λ_d denote its GRPO span weight. The token-level advantage is

$$\hat{A}_{i,t} = \begin{cases} \mathcal{N}_G(r_i^{\text{overall}}) & \text{coupled judge,} \\ \lambda_d \mathcal{N}_G(r_{i,d}) & \text{decoupled judge, } t \in T_{i,d}, \\ 0 & \text{decoupled judge, } t \notin \bigcup_d T_{i,d}. \end{cases} \quad (6)$$

We set $\lambda_d = 1$ for all four core dimensions. Thus, the C1 reward is applied only to the C1 rationale span and cannot directly cancel the C4 reward in the token-level objective, although all dimensions still share model parameters. We maximize the clipped surrogate with a reverse-KL anchor to the

SFT reference,

$$\mathcal{J}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_i \frac{1}{|o_i|} \sum_t \min(\rho_{i,t} \hat{A}_{i,t}, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) \hat{A}_{i,t}) \right] - \beta \mathbb{D}_{\text{KL}}[\pi_\theta \| \pi_{\text{ref}}], \quad (7)$$

with $\rho_{i,t} = \pi_\theta(o_{i,t} \mid q, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$ and the unbiased low-variance k_3 estimator for the KL term. Starting from the SFT-merged checkpoint, both GRPO runs initialize a new rank-64 LoRA adapter and use $G = 8$, learning rate $1\text{e-}5$, $\beta = 0.04$, and three epochs. The rank-64 adapter is merged into the backbone after GRPO. A dimension whose G rollouts all receive the same reward has zero normalized advantage and contributes no policy gradient. We add no format reward.

4.5. Dimension-wise supervision

The teacher, SpeechJudge, and our coupled student show a tendency for per-dimension verdicts to follow the overall winner. This motivates testing whether separately generated labels and removal of the shared overall-verdict target can produce more varied dimension-wise outputs. We split the rubric into single-dimension prompts and query the teacher *four times per dimension*, each call producing a CoT and score pair for that dimension only. Because repeating this procedure for every dimension would multiply annotation cost, we construct the Decouple-Live-SpeechJudge data using only the four core dimensions C1–C4; L1–L3 are omitted from both the training targets and model outputs. Per dimension we keep 4-0, 3-1, and three-agreeing-votes-plus-one-abstention outcomes and mask the rest. Consequently, an uncertain dimension contributes zero token loss in SFT and zero reward advantage in GRPO, while confident dimensions from the same pair remain trainable — the retention decision is made per (pair, dimension), not per pair. The surviving per-dimension responses are concatenated into one target rationale.

The student is trained and evaluated with one joint C1–C4 rubric prompt and emits one verdict per core dimension with no aggregate conclusion. In the name Decouple-Live-SpeechJudge, *Decouple* refers to removing the shared overall-verdict target and applying dimension-specific supervision and span-local rewards.

4.6. Adaptive audio attention bias

We also adopt the adaptive audio attention bias of [11], a mechanism targeting a plausible failure mode of long-CoT audio judgment: as the rationale grows, attention mass migrates to the already-generated text and the $\sim 12\text{k}$ -character rubric, and the audio itself is progressively ignored. We partition every token j into four regions by a deterministic rule — AUDIO (positions inside `<|audio_start|> ... <|audio_end|>`), COT (positions with label $\neq -100$,

i.e. the assistant rationale), PROMPT (everything before the `## The text content` marker that is not AUDIO, i.e. system prompt plus rubric), and BASE (the remainder: the target transcript and turn markers) — and apply a learnable per-head multiplicative gate on the attention logits, $s_{ij} \leftarrow s_{ij} + \log g(j)$, with

$$g(j) = \begin{cases} 1 + \sigma(f_{\text{audio}}) & j \in \text{AUDIO} \\ 2\sigma(f_{\text{prompt}}) & j \in \text{PROMPT} \\ \sigma(f_{\text{cot}}) & j \in \text{COT} \\ 1 & j \in \text{BASE}. \end{cases} \quad (8)$$

BASE is ungated and fixed at 1 to anchor the scale; initializing $f = 0$ gives $g = (1.5, 1, 0.5, 1)$, i.e. a mild prior that up-weights audio and down-weights the rationale. A representative training instance has 2618 PROMPT, 426 AUDIO, 634 COT and 80 BASE tokens.

In our exploratory run, the mechanism trains stably but does not increase test-set accuracy, so it is not included in the final model or the main result tables. The run uses COT spans of ~ 600 – 700 tokens and audio spans of ~ 200 – 430 tokens; whether the mechanism becomes useful for substantially longer rationales remains an open question.

5. EXPERIMENTS

5.1. Setup

Live-SpeechJudge is initialized from Qwen3-Omni and trained with attention-only LoRA in bf16 using the SWIFT framework [18]. All training runs use a fixed random seed and are run on 4 NVIDIA 5000-series GPUs with 72GB memory each. Coupled GRPO uses batched vLLM inference, whereas Decouple-Live-SpeechJudge GRPO uses native Hugging Face generation. The Gemini annotation used for distillation costs approximately RMB 10,000 in total. We compare against SpeechJudge-GRM [6] and Gemini-3.1-pro-preview under the same seven-dimension prompt. Gemini produces one judgment per test pair in the presented order without order swapping, reflecting its intended single-call use and the cost of repeated proprietary-model inference. SpeechJudge-GRM and our students each produce ten sampled judgments per pair, five in each A/B presentation order. Swapped score pairs are mapped back to the physical clips, the scores for each clip are averaged across the ten judgments, and the final verdict is the sign of the resulting mean score difference. To expose the contribution of inference-time sampling, we also report a 1-sample result for our final checkpoints: this uses the score pair from the first sampled judgment, without skipping to a later sample or resampling. An equal-score tie remains a tie and is counted as an error against the A/B human label. Gemini and each student 1-sample row use exactly one generated judgment per test pair. The 10-sample rows additionally benefit from repeated sampling and balanced order. We report final-verdict agreement with the human label

Table 2. Overall accuracy (%) on four independent high-expressiveness test sets and the T1★ unanimous subset. Gemini uses one call. For our models, “1 sample” uses the first sampled judgment, whereas “10-sample mean” averages scores from five judgments in each A/B presentation order. v1 is Qwen3-Omni + two-stage SFT + dimension weights; v1+GRPO adds GRPO post-training. v0 is a 10-sample confidence-threshold ablation trained on 2300 pairs retained at either 4-0 or 3-1.

Model / test set	T1 (417)	T1★ (119)	T2 (111)	T3 (315)	T4 (200)
SpeechJudge-GRM [6]	233/417 = 55.88	86/119 = 72.27	54/111 = 48.65	170/315 = 53.97	84/200 = 42.00
Gemini-3.1-pro-preview (1 sample) [12]	272/417 = 65.23	86/119 = 72.27	73/111 = 65.8	231/315 = 73.3	116/200 = 58.0
w/o stage-1 SFT (10-sample mean)	221/417 = 53.0	86/119 = 72.3	67/111 = 60.4	220/315 = 69.9	91/200 = 45.5
single-stage SFT, all data pairs (10-sample mean)	303/417 = 72.7	96/119 = 80.67	75/111 = 67.6	252/315 = 80.0	149/200 = 74.5
Live-SpeechJudge v0, 4-0 + 3-1 (10-sample mean)	297/417 = 71.22	94/119 = 78.99	73/111 = 65.77	254/315 = 80.63	146/200 = 73.0
Live-SpeechJudge v1 (1 sample)	271/417 = 64.99	89/119 = 74.79	70/111 = 63.06	233/315 = 73.97	133/200 = 66.50
Live-SpeechJudge v1 (10-sample mean)	295/417 = 70.74	96/119 = 80.67	78/111 = 70.27	259/315 = 82.22	160/200 = 80.00
Live-SpeechJudge v1 + GRPO (1 sample)	283/417 = 67.87	99/119 = 83.19	73/111 = 65.77	255/315 = 80.95	149/200 = 74.50
Live-SpeechJudge v1 + GRPO (10-sample mean)	297/417 = 71.22	98/119 = 82.35	78/111 = 70.27	261/315 = 82.86	167/200 = 83.5

as the evaluation metric. For paired comparisons on the same test pairs, we estimate sampling uncertainty using 10,000 paired bootstrap resamples. Each resample draws test-pair indices with replacement and applies the same indices to both systems; the 2.5th and 97.5th percentiles of the resulting accuracy-difference distribution form the 95% confidence interval. Unless otherwise stated, training results are from one fixed-seed run and accuracy differences are descriptive point estimates rather than multi-seed estimates.

5.2. Main results

Five observations follow from Table 2.

1. Under 10-sample balanced-order inference, the final GRPO model has higher point accuracy than a single Gemini judgment in every reported column. This is a stronger inference operating point that uses repeated sampling and order balancing, rather than a compute-matched comparison with Gemini.
2. Ten-sample aggregation usually improves agreement. Relative to one sample, the final GRPO model gains 3.35, 4.50, 1.91, and 9.00 points on T1, T2, T3, and T4, respectively; T1★ instead decreases by 0.84 points. This comparison reflects the combined effect of repeated sampling and A/B order balancing rather than either factor in isolation.
3. At the 10-sample operating point, the descriptive change from v1 SFT to v1+GRPO is +0.48, +1.68, 0.00, +0.63, and +3.50 points on T1, T1★, T2, T3, and T4, respectively. The measured contribution therefore varies substantially across test sets.
4. The stricter filtering configuration performs better in four of five columns despite using fewer pairs. Relative to v1’s 4-0-only supervision, v0 admits 3-1 outcomes and grows to 2300 pairs, but its 10-sample accuracy is lower by 1.68 points on T1★, 4.50 on T2, 1.59 on T3, and 7.00 on T4; T1 is higher by 0.48 points. This ablation favors 4-0-only filtering in the present training configuration, but does not separate label confidence from the accompanying change in dataset

Table 3. Paired-bootstrap comparison of 10-sample Live-SpeechJudge v1+GRPO against v1 SFT. Differences are in percentage points; confidence intervals use 10,000 paired resamples.

Test set	GRPO – SFT	95% CI
T1	+0.48	[−2.64, +3.60]
T1★	+1.68	[−2.52, +6.72]
T2	0.00	[−7.21, +7.21]
T3 pooled	+0.63	[−1.90, +3.18]
T4	+3.50	[−1.50, +8.50]

composition.

5. Relative to the 10-sample v1 SFT row, removing the qwen3-TTS-GDPO-vs.-human stage is associated with a 34.5-point lower T4 estimate (45.5%) in the human-vs.-TTS regime. This is consistent with, but does not prove, large quality-gap examples helping establish basic prosody-judgment behavior before fine-grained training. With the same data merged into one stage, accuracy is lower than sequential training on T2, T3, and T4, higher on T1, and equal on T1★. These single-run results motivate the sequential schedule but do not isolate curriculum order from training dynamics.

All intervals in Table 3 include zero. We therefore treat the 10-sample SFT-to-GRPO changes as descriptive point estimates rather than statistically significant improvements under test-pair sampling uncertainty. T1★ is a subset of T1 and is not treated as independent evidence.

5.3. Position bias

SpeechJudge-GRM assigns the second-presented clip nearly one additional point on average and selects B in 66.8% of judgments under this protocol. By contrast, Live-SpeechJudge has a measured B–A gap of only +0.012 and a 50.7% B win rate (Table 4). This smaller measured gap is consistent with our position-balanced supervision: after votes are mapped

Table 4. Position statistics on the 417-pair T1 test set (qwen3-TTS base vs. SFT). Each physical pair is scored 10 times with balanced A/B presentation order, yielding 4170 judgments per model. Mean A/B are the reported 1–10 overall-section scores for the presentation slots after pooling both physical clip assignments; t is computed from pair-level mean slot differences.

Model	Mean A	Mean B	B–A	t	B win
Live-SpeechJudge	7.842	7.854	+0.012	+0.4	50.7%
SpeechJudge-GRM	7.061	8.039	+0.978	+11.7	66.8%

back to the physical clips, retained pairs are assigned to presentation orders so that the winner appears equally often in position A and position B.

5.4. Dimension-wise supervision and output diversity

D-LSJ removes the shared overall-verdict target and applies dimension-specific supervision and span-local rewards. Across the evaluation sets, 24.3–60.6% of its two-stage-SFT outputs contain at least two different non-tie core-dimension verdicts, demonstrating the capacity to issue different dimension verdicts. As a teacher-defined cross-dimension stress test, we further select 140 audio pairs for which all four independently queried teacher labels are confident (4-0, 3-1, or three agreeing non-tie votes plus one abstention) but disagree across dimensions. On this same set, the coupled LSJ produces no non-unanimous core-verdict vector (0/140) and obtains 48.0% per-dimension agreement with the teacher. D-LSJ after SFT produces 90/140 non-unanimous outputs and 380/560 (67.9%) per-dimension agreement; span-local GRPO further changes these results to 100/140 (71.4%) and 400/560 (71.4%), respectively. Because this stress set is selected by disagreement among independently queried teacher labels, it diagnoses output diversity and teacher alignment rather than human-grounded per-dimension correctness. Within that scope, the same-set comparison associates the decoupled training design with fewer unanimous verdict vectors and higher agreement with the teacher labels.

D-LSJ also allows weighted aggregation of its dimension verdicts. We reserve a held-out validation split from the training pool exclusively for selecting the aggregation weights, choose $w = (3, 0.5, 0.5, 3)$ on that split, and freeze it before evaluating on any test set. Table 5 shows that this validation-selected scheme, which emphasizes C1 and C4, has a 4.3-point higher T1 estimate than uniform C1–C4 weights. In the descriptive cross-configuration comparison against the coupled LSJ+GRPO row, D-LSJ after two-stage SFT has higher point accuracy on T1, T4, and T2, lower accuracy on T1*, and slightly lower accuracy when T3a and T3b are pooled. Because the training objectives and stages differ, this comparison does not isolate the effect of decoupling. The remaining SFT rows are fixed-weight ablations rather than candidates

Table 5. Dimension-wise student after two-stage SFT, aggregated under different inference weight vectors w_d over the four core dimensions in the order C1/C2/C3/C4 (%). The 3/.5/.5/3 scheme is selected on a held-out validation split and frozen before test evaluation; the other SFT schemes are fixed-weight ablations. The aggregate verdict is the sign of $\sum_d w_d(s_d^A - s_d^B)$. Columns are T1 (417 pairs), T1* (119), T4 (200), T3a (166), T3b (149) and T2 (111). The first three rows use coupled-judge outputs, for which T3 is reported as a single 315-pair number.

Weight scheme	T1	T1*	T4	T3a	T3b	T2
LSJ v1 + GRPO (coupled)	71.2	82.4	83.5	82.9	70.3	
Gemini (coupled prompt)	65.2	72.3	58.0	73.3	65.8	
Gemini, 3/2/1/1	66.7	81.5	75.5	75.2	60.4	
<i>Decouple-Live-SpeechJudge (two-stage SFT)</i>						
Uniform (C1–C4)	69.3	78.2	78.5	73.5	83.9	71.2
3/2/1/1	69.1	76.5	77.5	74.7	80.5	71.2
3/.5/.5/3	73.6	80.7	85.5	77.7	86.6	71.2
C2C3C4 only	68.4	76.5	79.5	74.1	82.6	70.3
C4 only	69.3	79.0	79.5	75.3	85.2	73.0

Table 6. Point agreement with high-confidence human labels (%). Gemini uses one call; D-LSJ uses the first sampled judgment (1-sample) or the balanced-order mean (10-sample). Parentheses give set sizes.

Model / inference	C1 (224)	C2 (172)	C3 (142)	C4 (160)	Total (698)
Gemini (1-sample)	74.55	77.33	80.28	84.38	78.65
D-LSJ (1-sample)	68.75	65.70	76.06	86.88	73.64
D-LSJ (10-sample)	79.02	80.81	85.92	93.13	84.10
D-LSJ + Agg.-GRPO (1-sample)	70.54	71.51	79.58	90.62	77.22
D-LSJ + Agg.-GRPO (10-sample)	79.91	83.14	87.32	92.50	85.10
D-LSJ + Span-GRPO (1-sample)	71.88	76.74	76.76	88.75	77.94
D-LSJ + Span-GRPO (10-sample)	83.04	83.14	85.21	94.38	86.10

selected from the test results. We evaluate the effect of per-dimension GRPO only against human per-dimension labels in Section 5.5.

5.5. Human-annotated per-dimension evaluation

The four evaluation sets isolate one core rubric dimension each and test whether a judge recovers a high-confidence human label for that target dimension. We use separate dimension-specific sets because, for a given audio pair, a confident human preference in one dimension does not imply confident preferences in all four; requiring a complete C1–C4 verdict vector would either discard reliable single-dimension labels or force uncertain dimensions into the evaluation. These sets therefore test human agreement for individual dimensions rather than full C1–C4 verdict vectors on the same audio pairs. The same three trained annotation staff independently label each pair using the corresponding dimension section of the Gemini prompt. Every retained pair has a unanimous 3-0, non-tie label for its target dimension; majority-with-abstention cases are excluded.

In this fixed-seed comparison, aggregate-reward GRPO

Table 7. Paired-bootstrap comparisons for 10-sample per-dimension agreement. Entries report the point-accuracy difference in percentage points and its 95% confidence interval from 10,000 paired resamples.

Dimension	Span-GRPO — SFT	Span-GRPO — Agg.-GRPO
C1	+4.02 [−0.89, +9.38]	+3.12 [−1.79, +7.59]
C2	+2.33 [−2.33, +6.98]	0.00 [−5.23, +5.23]
C3	−0.70 [−4.23, +2.82]	−2.11 [−6.34, +2.11]
C4	+1.25 [−1.88, +4.38]	+1.88 [−1.88, +5.62]

changes pooled agreement over SFT from 73.64% to 77.22% with one sample and from 84.10% to 85.10% with 10 samples. Span-local GRPO has pooled point estimates of 77.94% and 86.10%, respectively, 0.72 and 1.00 points above the shared aggregate-advantage run.

At 10 samples, Span-GRPO has higher point estimates than SFT on C1, C2, and C4 and a lower estimate on C3. Relative to aggregate-reward GRPO, it is higher on C1 and C4, equal on C2, and lower on C3. All per-dimension bootstrap intervals in Table 7 include zero, so we treat these differences as descriptive rather than statistically significant. The bootstrap intervals quantify test-pair sampling uncertainty for this fixed-seed run and do not estimate variation across training seeds.

5.6. Best-of-N selection

We additionally deploy Live-SpeechJudge as the decision component of a Best-of-8 inference pipeline. For each transcript, the TTS model samples eight candidate utterances. Live-SpeechJudge evaluates all $\binom{8}{2} = 28$ distinct candidate pairings, and the selected utterance is the candidate with the largest number of pairwise wins,

$$\hat{a} = \arg \max_{a_i} \sum_{j \neq i} \mathbf{1}[V(a_i, a_j) = a_i]. \quad (9)$$

This tournament-style aggregation requires no absolute quality calibration and uses the same pairwise judgment task on which the model is trained. Professionally trained contractor annotators evaluate 400 eight-candidate sets. For a high-confidence evaluation, we retain the 136 sets in which two annotators produce identical ordered top-3 rankings, following the use of unanimous evaluation labels in SpeechJudge-Eval [6]. The remaining 264 sets are excluded because the perceptual differences among their candidates are not sufficiently distinct to support a confident ordered top-3 annotation; this confidence-based inclusion decision is made without reference to model predictions. Hit@ k indicates that the tournament winner lies within this agreed human top- k set. On these 136 retained sets, the selector obtains Hit@1, Hit@2, and Hit@3 rates of 98/136 = 72.06%, 106/136 = 77.94%, and 116/136 = 85.29%, respectively, compared with random-selection baselines of 12.5%, 25.0%, and 37.5%. These rates

characterize the high-confidence test set and are not estimates over all 400 collected sets. Qualitative examples on our demo page additionally illustrate the serving workflow and compare the selected utterance with the other seven candidates.

5.7. Scope of the evidence

Our results have four boundaries. First, training comparisons use one fixed seed, so accuracy differences are descriptive point estimates rather than estimates of run-to-run variation. The paired-bootstrap intervals quantify uncertainty from the sampled test pairs, not variation across training seeds. All intervals for the coupled judge’s 10-sample SFT-to-GRPO comparison, as well as all reported per-dimension intervals for Span-GRPO versus SFT and aggregate-reward GRPO, include zero. Second, the 10-sample rows use more inference and balanced presentation order than the single-call Gemini baseline; they characterize a stronger operating point rather than a compute-matched comparison. Third, the 140-pair cross-dimension stress set is defined by independently queried teacher labels, whereas the human per-dimension sets evaluate one target dimension per audio pair. Finally, the Best-of-8 rates apply to the 136 high-confidence sets retained from 400 collected sets. D-LSJ is limited to C1–C4, and using its vector outputs to optimize the TTS backbone remains future work.

6. CONCLUSION

We presented Live-SpeechJudge, a pairwise multi-dimensional prosody judge for e-commerce livestream TTS. Its 10-sample balanced-order aggregation has higher point accuracy than one Gemini judgment on all reported columns and shows a small presentation-slot gap. This comparison uses a stronger inference operating point rather than matched inference compute. The paired-bootstrap intervals for the coupled judge’s 10-sample SFT-to-GRPO changes all include zero, so those changes remain descriptive. Decouple-Live-SpeechJudge removes the overall-verdict target and applies dimension-specific supervision and span-local rewards to C1–C4. Its non-unanimous outputs demonstrate varied dimension verdicts. In one fixed-seed run, Span-GRPO changes pooled human agreement from 73.64% to 77.94% with one sample and from 84.10% to 86.10% with 10 samples, although all per-dimension confidence intervals include zero. On the retained 136-set Best-of-8 evaluation, the selected candidate achieves Hit@1/2/3 rates of 72.06%/77.94%/85.29%. Future work will use D-LSJ’s dimension-wise judgments to optimize the TTS backbone.

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